

Surface-bulk hybridization enhances the Berry curvature dipole in GaAs(110)

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Symmetry breaking at crystalline boundaries can enable electronic responses that are forbidden in the bulk. Here, we investigate the Berry curvature dipole (BCD) at GaAs(110) surfaces using first-principles calculations. While the BCD vanishes in bulk GaAs due to symmetry constraints, the reduced surface symmetry permits a finite dipole. We show that surface relaxation promotes hybridization between bulk and surface states, which strongly influences the surface BCD. Specifically, in unrelaxed slabs, the BCD arises predominantly from interband coupling among exponentially localized surface states lying within the projected bulk band gap. Surface relaxation shifts these states into the energy range of the bulk continuum, transforming them into surface resonances and promoting hybridization with bulk-like states. This hybridization generates pronounced Berry-curvature hot spots, resulting in an overall enhancement of the BCD. Our results highlight the importance of surface relaxation for accurately describing surface-bulk hybridization, which in turn plays a key role in Berry-curvature-driven responses at crystalline boundaries.

I. INTRODUCTION

In crystalline materials, several electronic properties are linked to the geometric and topological properties of the electronic band structure [1–14]. Starting from the quantum geometric tensor, different transport and optical phenomena find a source in either the real part of this tensor, i.e., the quantum metric, or its imaginary part, known as the Berry curvature (BC).

The BC can give rise to the so-called anomalous Hall effect, which produces a Hall current linear in the applied electric field, provided that time-reversal symmetry $\mathcal{T}$ is broken. In contrast, in $\mathcal{T}$ -invariant noncentrosymmetric materials, a finite BC is allowed, but the linear Hall response vanishes. In this case, the leading contribution to the Hall response arises at second order in the electric field [15]. The so-called nonlinear Hall effect is directly linked to the first moment of the BC, known as the Berry curvature dipole (BCD).

Since the BCD is strongly constrained by the symmetries of the underlying point group, low-symmetry materials are natural candidates for hosting nonlinear Hall effects in the bulk [16–22]. Moreover, surfaces may naturally break symmetries, allowing for a finite BCD even when it is forbidden in the bulk [23–27]. Furthermore, while the origin of the BC is typically associated with internal degrees of freedom, such as spin, orbital, or site degrees of freedom, the presence of a surface introduces an additional source of BC through the dependence of the penetration depth of the Bloch wave function on the conserved crystalline momenta. For example, in models of Weyl semimetals, the delocalization of surface states conforming the Fermi arcs has been shown to lead to divergences of the BC, thereby enhancing the BCD as well [23, 25].

Gallium arsenide (GaAs) is a prototypical III–V semiconductor with direct relevance for electronic devices [28–

33], whose bulk phase crystallizes in space group 216 (zincblende structure) [34]. Although this structure lacks inversion symmetry, the BCD is forbidden by the combined action of its space-group symmetries [35]. At the same time, GaAs and related III–V compounds are well suited for thin-film and surface studies, as they can be reliably cleaved along several crystallographic directions. Moreover, it can be grown and carefully controlled by Molecular Beam Epitaxy and allows the gating of heterostructures and quantum wells with 2D electron gasses (2DEG). In particular, the (110) surface is of special interest, since the symmetry is reduced to space group 31, resulting in a noncentrosymmetric structure with lower symmetry that allows for a finite BCD.

The GaAs (110) surface has been extensively investigated both experimentally and theoretically [36–49], and its electronic structure is known to be strongly affected by surface relaxation. In particular, relaxation can induce substantial changes in the localization of surface states [44]. These effects are expected to have a significant impact on band-geometric properties such as the BCD and the Berry curvature [23].

In this work, we systematically investigate the electronic structure and Berry curvature dipole of GaAs(110) slabs of varying thickness, focusing on the role of surface states and structural relaxation. Using first-principles calculations, we analyze how symmetry and surface relaxation govern nonlinear transport responses in this prototypical semiconductor. Specifically, we show that the well-known reconstruction of the GaAs(110) surface strongly enhances surface–bulk hybridization, which in turn has a profound impact on the surface BCD.

This article is organized as follows. In Section II, we describe the methods used throughout the work. In Section III, we study the electronic band structure of GaAs(110) with and without surface relaxation. In Section IV, we present calculations of the the BCD in GaAs,

focusing on the effects of the surface relaxation. Lastly, in Section V, we present our concluding remarks.

II. METHODS

A. Density-functional calculations

To explore the physics of the (110) surface, we investigated the electronic band structure of GaAs slabs with surfaces perpendicular to the bulk (110) direction. The natural unit cell for constructing such slabs is no longer cubic but tetragonal, with lattice parameters satisfying $a \neq b = c$, where c is perpendicular to the (110) surface [50]. The finite slabs can be constructed such that they belong to space group $Pmn2_1$. The corresponding crystal structures are shown in Fig. 1(a) and (b). Although the symmetry is reduced relative to the bulk, the slab space group still contains a mirror plane, illustrated in Fig. 1(c), which constrains the BCD to lie perpendicular to it.

We constructed supercells with slab thicknesses $t = nc$, where $n = 6, 8, 10$, and 12 . A vacuum region was introduced along the c direction to decouple periodic images of the slab. A vacuum thickness of $15 \text{ \AA}$ was found to be sufficient to suppress spurious interactions between adjacent slabs. The use of a larger vacuum leads to differences in the total energy of the order of 10^{-9} eV .

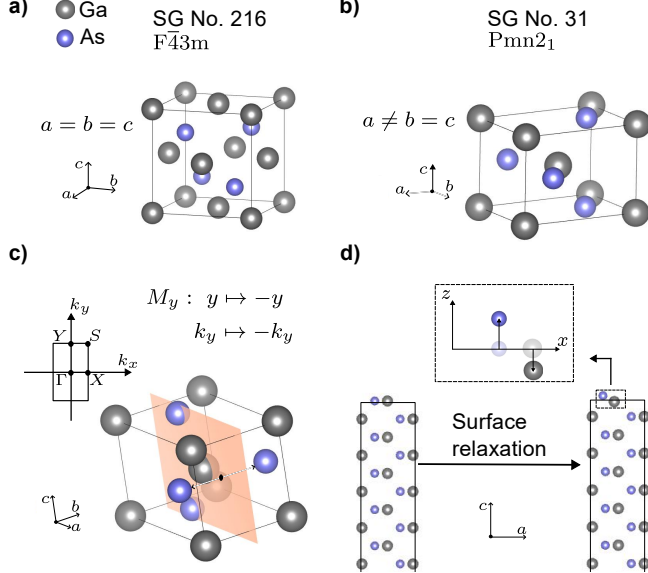


FIG. 1. Structural and symmetry properties of bulk and (110)-oriented GaAs. (a) Crystal structure and space group of bulk GaAs. (b) Crystal structure and space group of the GaAs(110) surface. (c) Brillouin zone and mirror symmetry (M_y) preserved at the GaAs(110) surface. (d) Atomic displacement induced by surface relaxation.

We performed density functional theory (DFT) calculations using the FPLO package [51, 52] (version 22.02)

within the generalized gradient approximation (GGA). For the self-consistent calculation of the charge density, we employed a $\mathbf{k}$ -mesh of $12 \times 12 \times 1$ subdivisions, which was found to be sufficient to converge the total energy.

To assess the effects of surface relaxation on the electronic structure, structural optimizations were performed for the first surface layer. The atomic positions were relaxed by minimizing the forces within the scalar relativistic approximation with a tolerance of $1 \times 10^{-3} \text{ eV/\AA}$. Subsequent electronic structure calculations were performed with the fully-relativistic four-component scheme implemented in FPLO. We have checked that relaxing deeper layers only introduce moderate corrections to the surface atomic reconstruction.

Structurally, the relaxation results in the As atom being displaced outward from the slab, while the Ga atom is shifted toward the interior of the slab. This is illustrated schematically in Fig. 1(d). According to our calculations, the angle of rotation of the surface Ga-As bond is of approximately 28° , consistent with what is reported in the literature for this material [53–55]. The repository [56] provides the different structural models used throughout this work.

For each slab thickness, tight-binding models were constructed using symmetry-conserving projected Wannier functions based on the Ga and As $4s$ and $4p$ orbitals, as implemented in pyFPLO [57]. These models accurately reproduce the low-energy electronic structure obtained from DFT, with a calculated standard deviation of approximately 6 meV in the energy window ranging from -1 to 1 eV . These models were used to compute the BC and perform further postprocessing analyses of the band structure.

B. Berry curvature dipole

The electronic structure is described by the Wannier Hamiltonian $H(\mathbf{k})$ obtained from the projective Wannier construction described above. Its eigenstates satisfy

$$H(\mathbf{k}) |n, \mathbf{k}\rangle = \varepsilon_n(\mathbf{k}) |n, \mathbf{k}\rangle, \quad (1)$$

where $|n, \mathbf{k}\rangle$ and $\varepsilon_n(\mathbf{k})$ denote, respectively, the Bloch eigenstate and energy of band n at crystalline momentum $\mathbf{k}$. The velocity operators are defined as

$$v_a(\mathbf{k}) = \frac{1}{\hbar} \frac{\partial H(\mathbf{k})}{\partial k_a},$$

where, owing to the two-dimensional periodicity of the slab geometry, $a = x, y$.

The Berry curvature is a band-geometric quantity that characterizes the local geometry of the Bloch wave functions in momentum space. For the n th band, its z component can be expressed as

$$\Omega_n^z(\mathbf{k}) = -\text{Im} \sum_{m \neq n} \frac{\langle n, \mathbf{k} | v_x | m, \mathbf{k} \rangle \langle m, \mathbf{k} | v_y | n, \mathbf{k} \rangle}{[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2}, \quad (2)$$

where the sum runs over all bands $m \neq n$ at the same crystalline momentum $\mathbf{k}$.

The Berry curvature dipole (BCD) is a pseudotensor that characterizes the first moment of the BC in momentum space [15]. It is defined as

$$D_{ab} = \sum_n \int \frac{d^d k}{(2\pi)^d} f_0(\varepsilon_n) \partial_a \Omega_n^b(\mathbf{k}), \quad (3)$$

where d denotes the dimensionality of the system, f_0 is the Fermi–Dirac distribution, and $\partial_a \equiv \partial/\partial k_a$. Introducing the shorthand

$$\int_{\mathbf{k}} \equiv \sum_n \int \frac{d^d k}{(2\pi)^d}, \quad (4)$$

the BCD can be written compactly as

$$D_{ab} = \int_{\mathbf{k}} f_0 \partial_a \Omega_b. \quad (5)$$

Integrating by parts yields

$$D_{ab} = - \int_{\mathbf{k}} (\partial_a f_0) \Omega_b, \quad (6)$$

which highlights the role of states near the Fermi surface. Indeed, $\partial_a f_0 = (\partial f_0/\partial \varepsilon_n) \partial_a \varepsilon_n$, and in the zero-temperature limit $-\partial f_0/\partial \varepsilon_n$ approaches a Dirac delta function centered at the Fermi energy.

To analyze the spatial distribution of the BCD across a slab, we implemented a Wannier projection scheme for the BCD. At each $\mathbf{k}$ point, the Berry curvature of the Hamiltonian eigenstates was evaluated together with the velocity matrix elements obtained from derivatives of the Wannier Hamiltonian. The contribution of each Wannier function i to the BCD was then obtained by projecting the band-resolved BCD integrand onto the Wannier basis:

$$D_{ab}^i = \int_{\mathbf{k}} \sum_{n,j} \Omega_{b,n}(\mathbf{k}) \frac{\partial f_0}{\partial \varepsilon_n} C_{i,n}^*(\mathbf{k}) v_{a,ij}(\mathbf{k}) C_{j,n}(\mathbf{k}). \quad (7)$$

Here, $C_{i,n}(\mathbf{k})$ are the eigenstate coefficients in the Wannier basis, $v_{a,ij}(\mathbf{k})$ denotes the velocity matrix element, and i labels the Wannier function onto which the BCD contribution is projected. This decomposition provides an orbital- and site-resolved representation of the BCD, enabling the identification of the microscopic regions responsible for the nonlinear Hall response. By assigning each Wannier orbital to its corresponding atomic layer, we further obtain a layer-resolved decomposition of the BCD.

To identify the dominant interband processes contributing to the BCD, we further decompose the BC of each band into contributions arising from all other bands. From Eq. 2, we define

$$\Omega_n^z(\mathbf{k}) = \sum_{m \neq n} \Omega_{nm}^z(\mathbf{k}), \quad (8)$$

where

$$\Omega_{nm}^z(\mathbf{k}) = -\text{Im} \frac{\langle n, \mathbf{k} | v_x | m, \mathbf{k} \rangle \langle m, \mathbf{k} | v_y | n, \mathbf{k} \rangle}{[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2} \quad (9)$$

denotes the contribution to the BC of band n arising from band m . This decomposition allows us to express the contribution of band n to the BCD as

$$D_{ab}^n = - \sum_{m \neq n} \int_{\mathbf{k}} \partial_a f_0 \Omega_{nm}^b = \sum_{m \neq n} D_{ab}^{n \leftarrow m}. \quad (10)$$

Here, $D_{ab}^{n \leftarrow m}$ denotes the contribution to the BCD of band n arising from the interband contribution of band m .

Finally, for the numerical evaluation of the BCD, we used a 2000×2000 $\mathbf{k}$ -point mesh. The integration was restricted to the first quadrant of the Brillouin zone, exploiting crystal and time-reversal symmetries to reconstruct the contributions from the remaining regions.

III. GaAs(110) BAND STRUCTURE

The electronic band structure of GaAs has been extensively studied, both theoretically and experimentally. GaAs is a semiconductor with an experimental direct band gap at Γ of approximately 1.4 eV [34]. The (110) surface is known to host electronic states within the bulk band gap, which is a characteristic feature of its surface electronic structure. To investigate these states, we computed the electronic band structure of GaAs slabs of varying thickness. As shown in Fig. 2, a set of states emerges above and below the Fermi level for all thicknesses considered. Importantly, the number of these states does not scale extensively with the number of layers n : only four of such states are observed for all values of n . This thickness-independent number of states suggests that they are associated with the surfaces of the slab.

For $n = 12$, Fig. 2 also shows the projected bulk band structure, obtained by varying the crystalline momentum along the direction perpendicular to the surface (110). In the limit $n \rightarrow \infty$, it is expected that a portion of the spectral weight of the slab converges to the continuum formed by the projected bulk states. Already for $n = 12$, most of the slab band structure lies within the projected bulk bands, whereas the emergent states identified above remain within the bulk band gap.

As a quantitative measure of the spatial localization of a state, we introduce the surface polarization parameter

$$\eta(\mathbf{k}) = w_s(\mathbf{k}) - w_b(\mathbf{k}) = 2w_s(\mathbf{k}) - 1, \quad (11)$$

where $w_s(\mathbf{k})$ is the weight of a Bloch state on sites located near the surfaces of the slab, and $w_b(\mathbf{k})$ is the corresponding weight on intermediate, bulk-like sites. Owing to the normalization of the eigenstates, $w_s(\mathbf{k}) + w_b(\mathbf{k}) = 1$ for all $\mathbf{k}$ points in the Brillouin zone. With this definition, states

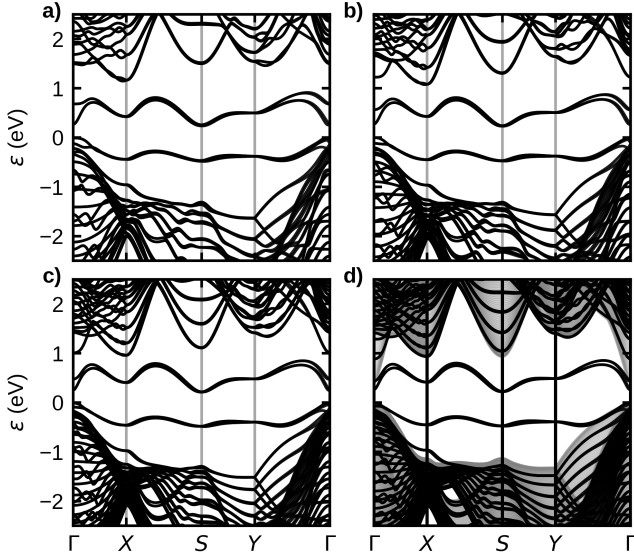


FIG. 2. Electronic band structure of GaAs(110) slabs. (a-d) Electronic band structures for slab thicknesses $n = 6, 8, 10$, and 12 , respectively, where n denotes the number of bulk unit cells forming the slab. In (d), the grey continuum spectrum corresponds to the bulk projection.

fully localized at the surfaces have $\eta(\mathbf{k}) = 1$, whereas states fully localized in the bulk have $\eta(\mathbf{k}) = -1$. The surface region is defined as the first three atomic layers at both the top and bottom surfaces of the slab.

Fig. 3(a) presents the band structure of the $n = 10$ slab, colored according to $\eta(\mathbf{k})$. The states lying within the band gap are strongly localized at the surfaces, with those in the immediate vicinity of the Γ point, near the top of the bulk valence band, exhibiting a somewhat larger spread into the bulk. Throughout this article, we use the notation g_{ci} ($i = 1, 2, 3, 4$) to denote the unoccupied states within the bulk band gap, and g_{vi} to denote the corresponding occupied states. The remaining conduction and valence bands are indexed by c_i (counted from the bottom of the conduction band) and v_i (counted from the top of the valence band), respectively. Hereafter, bands g_{ci} and g_{vi} are indicated by solid black lines to distinguish them from the other bands.

We next consider the effects of surface relaxation, implemented as described in Section II. The structural changes lead to significant modifications of the electronic band structure. In particular, the states that previously lay within the bulk band gap shift toward the energy range of the bulk states, as shown in Fig. 3(b). Due to their shift in energy, at selected $\mathbf{k}$ points, these states strongly hybridize with the bulk continuum and therefore extend into the bulk. In the rest of this paper, we explain how this change affects BC-based properties.

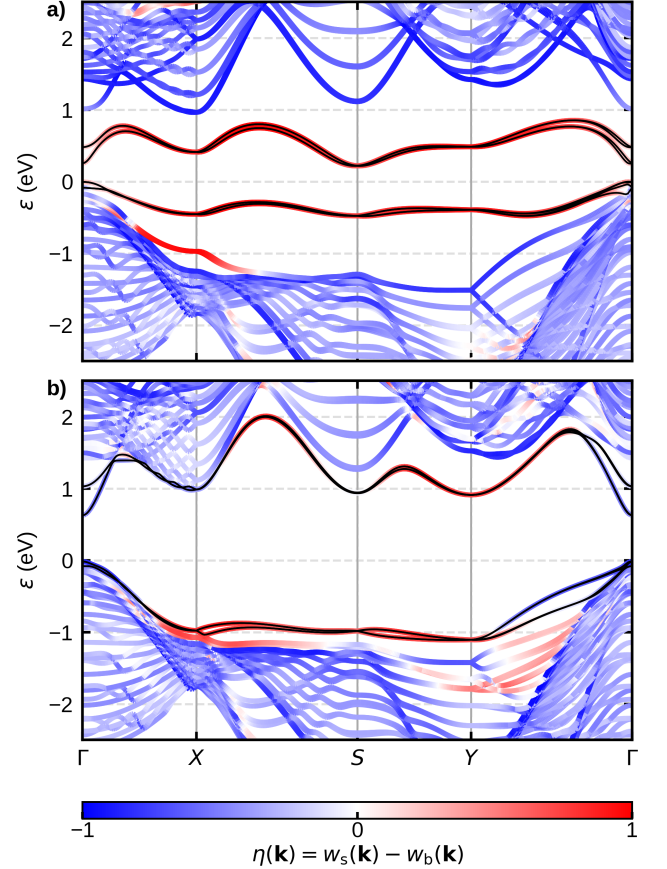


FIG. 3. Band structure along the high-symmetry path for the $n = 10$ slab. The colorbar indicates the surface polarization for (a) the slab without relaxation and (b) the slab with relaxed surface. The solid black lines highlight the states labeled g_{ci} and g_{vi} (see main text), which in (a) lie within the bulk band gap.

IV. BERRY CURVATURE DIPOLE IN GaAs

A. Bulk and Surface GaAs BCD

The BCD vanishes in bulk GaAs as a consequence of symmetry constraints. To determine which components of the BCD are allowed by symmetry, it is useful to recall that D^{ab} is a second-rank pseudotensor: the index a , associated with the momentum derivative ∂_a , transforms as a polar vector, whereas the index b , associated with the Berry curvature Ω_b , transforms as an axial vector. Consequently, the BCD transforms according to the direct product $\Gamma_V \otimes \Gamma_A$, where Γ_V and Γ_A denote the vector and axial-vector representations of the crystal point group. A finite BCD is possible only if this product contains the trivial irreducible representation A_1 , corresponding to a tensor component that remains invariant under all crystal symmetry operations.

For bulk GaAs, the point group is $\bar{4}3m$. The vector

representation corresponds to the three-dimensional irreducible representation (irrep) T_2 , whereas the axial-vector representation corresponds to T_1 . Their direct product decomposes as

$$\Gamma_V \otimes \Gamma_A = T_2 \otimes T_1 = A_2 \oplus E \oplus T_1 \oplus T_2.$$

Since this decomposition does not contain the trivial irrep A_1 , no component of the BCD remains invariant under all symmetry operations, and the BCD is therefore forbidden in bulk GaAs.

The situation changes at the (110) surface, where the symmetry is reduced to the point group $2mm$. In this case, the vector and axial-vector representations decompose as

$$\Gamma_V = A_1(x) \oplus B_1(y) \oplus B_2(z),$$

and

$$\Gamma_A = A_2(J_x) \oplus B_2(J_y) \oplus B_1(J_z),$$

respectively. Here, (x, y, z) denote the components of a polar vector, whereas (J_x, J_y, J_z) denote those of an axial vector. Since all irreducible representations of $2mm$ are one-dimensional, the product of a polar and an axial component transforms as the trivial representation only when both belong to the same irrep. Inspection of the above decompositions immediately shows that this occurs only for the pairs (y, J_z) and (z, J_y) , giving

$$A_1(y, J_z) \oplus A_1(z, J_y) \subset \Gamma_V \otimes \Gamma_A.$$

Accordingly, only the tensor components D^{yz} and D^{zy} are allowed by symmetry. For a two-dimensional slab, however, k_z is not a good quantum number, so only D^{yz} is physically relevant. Under these conditions, the BCD can be represented as an in-plane pseudovector with only D_y component allowed by symmetry.

Starting from the Wannier models of the slabs, we computed the D_y component of the BCD for both unrelaxed and relaxed surface structures. At this point, it is useful to discuss the quantitative impact of the band gap on our calculations. While the experimentally reported band gap of GaAs is approximately 1.4 eV [34], DFT typically underestimates its magnitude [58, 59], yielding significantly smaller values [60–65]. In principle, this discrepancy may affect the Berry curvature, which depends sensitively on the energy separation between bands, and consequently the magnitude of the BCD. However, the smallest interband gaps relevant to the features discussed below occur predominantly between conduction bands. We therefore expect the underestimated fundamental gap between the valence and conduction bands to play a less significant role in the qualitative behavior of the BCD considered here.

In addition, several other factors may affect the magnitude of the experimentally measured BCD and the associated nonlinear transport response, including disorder and the crystal domain structure. These factors make

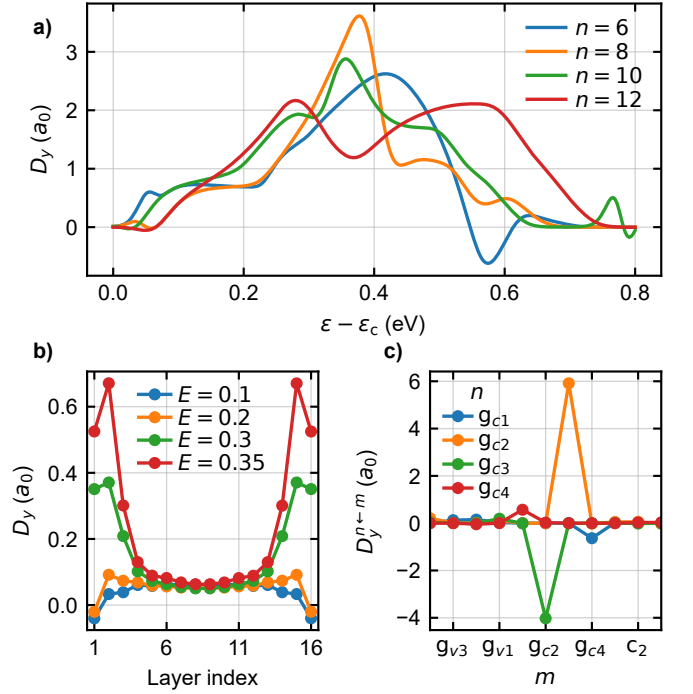


FIG. 4. Berry curvature dipole (BCD) for the unrelaxed slabs. (a) Energy dependence of the BCD with respect to the bottom of the conducting band. (b) Layer index dependence of the BCD for different energies with respect to ϵ_c . The unit for the energies is eV. (c) D_y^{n-m} at $\epsilon - \epsilon_c = 0.35$ eV. We include only the n that have a finite isoenergy surface at this energy, since those are the ones contributing to the BCD.

quantitatively accurate predictions of transport coefficients challenging. Since our analysis primarily focuses on the symmetry and microscopic origin of the BCD, we therefore present the results obtained directly from DFT and focus on a qualitative comparison of the changes induced by surface relaxation.

B. Unrelaxed Slabs

For unrelaxed slabs, we calculate the BCD within the energy window spanned by the conduction-band surface states, from 0 to 0.8 eV, as shown in Fig. 4(a) in units of Bohr radius a_0 . To analyze the real-space distribution of the BCD, we resolve its contributions into individual atomic layers at fixed energy following the procedure described in Sec. II. The layer-resolved contributions for the $n = 8$ slab are shown in Fig. 4(b). At energies close to the bottom of the conduction band ($\epsilon - \epsilon_c \lesssim 0.2$ eV), the BCD is predominantly concentrated in the intermediate, bulk-like layers. As the energy increases, a surface-dominated contribution emerges, primarily localized within the outermost three layers. The intermediate layers retain a finite contribution, which we attribute to finite-size effects. Indeed, in the bulk limit, $n \rightarrow \infty$, this contribution is expected to vanish by symmetry as the

full $\bar{4}3m$ point-group symmetry is recovered.

To understand the origin of the BC associated with the dominant surface contribution, we analyze the inter-band contributions to the BCD (Eq. 11), $D_y^{n \leftarrow m}$, shown in Fig. 4(c) for the $n = 8$ slab at $\varepsilon - \varepsilon_c \sim 0.35$ eV, where the BCD is predominantly localized at the surface. We find that the dominant contributions originate from inter-band matrix elements connecting surface-localized states, namely those lying within the bulk band gap. This analysis shows that the BCD is generated primarily by interband coupling between surface states, with negligible contributions from bulk-like bands.

Returning to the discussion of the underestimated band gap in DFT, we note that the contribution of a band m to the Berry curvature of band n scales as $1/[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2$. The dominant contributions identified here arise from small energy separations among surface states rather than from transitions across the fundamental valence-conduction band gap. We therefore expect the DFT underestimation of the latter to have a comparatively minor impact on the qualitative behavior of the BCD discussed here.

C. Relaxed Slabs

The strong band-structure reconstruction induced by surface relaxation is directly reflected in the BCD. As shown in Fig. 5(a), the BCD is strongly suppressed at low energies, between 0 and 0.3 eV, reaching values of only $\sim 0.66 a_0$. In this energy range, the conduction states retain a predominantly bulk-like character, as illustrated in Fig. 5(b), resulting in $D_y(\varepsilon)$ profiles that are nearly independent of slab thickness n , apart from an approximately rigid energy shift. At higher energies, the BCD increases significantly, reaching absolute values of 5–8 a_0 , approximately twice those obtained for unrelaxed slabs. This enhancement coincides with the onset of strong hybridization between surface-resonant states and an increasing number of bulk-like bands. Consequently, the detailed shape of $D_y(\varepsilon)$ becomes increasingly dependent on n , since both the number and energy distribution of the hybridizing bulk-like bands vary with slab thickness. Nevertheless, the overall magnitude of the dipole remains comparable for all values of n considered. We therefore expect the BCD profile to converge in the limit of sufficiently thick slabs, although reaching this regime is currently limited by the computational cost of the calculations.

We have verified that relaxing additional atomic layers does not qualitatively modify these conclusions. For energies below $\varepsilon - \varepsilon_c = 0.4$ eV, the BCD profiles closely resemble those shown in Fig. 5(a), apart from a small energy shift. At higher energies, however, the detailed profiles become increasingly dependent on the number of relaxed layers. As discussed above, we attribute this sensitivity to the large number of slab bands present in this energy window: even small modifications of the electronic

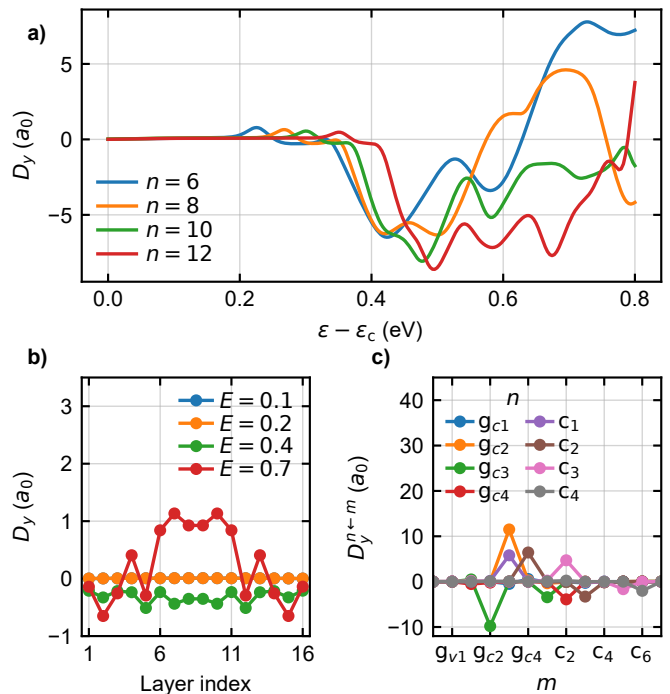


FIG. 5. Berry curvature dipole (BCD) for the relaxed slabs. (a) Energy dependence of the BCD with respect to the bottom of the conducting band. (b) Layer dependence of the BCD for different energies with respect to ε_c . The unit for the energies is eV. (c) $D_y^{n \leftarrow m}$ at $\varepsilon - \varepsilon_c = 0.7$ eV. We include only the n that have a finite isoenergy surface at this energy, since those are the ones contributing to the BCD.

structure can produce substantial changes in geometric quantities such as the BC and the BCD. This sensitivity is an inherent consequence of describing bulk continuum states within a finite slab geometry. Nevertheless, the overall magnitude of the BCD remains comparable for all relaxation depths considered, indicating that the main physical conclusions are robust against the number of relaxed layers.

The layer-resolved BCD shown in Fig. 5(b) also exhibits pronounced differences from the unrelaxed case. While in unrelaxed slabs the largest contributions originate from the surface layers, after relaxation the BCD is distributed throughout the entire slab. Consistently, the interband contributions to the BCD are also distributed among a broader set of bands than in the unrelaxed case, as shown in Fig. 5(c).

D. Hybridization of surface and bulk states

To analyze the momentum-space origin of the BCD across the entire BZ, we study the $\mathbf{k}$ -resolved density $D_y(\mathbf{k})$, which corresponds to the integrand in equation 6, for the $n = 8$ slab. At $\varepsilon - \varepsilon_c = 0.7$ eV, where D_y reaches its largest value, Fig. 6(a) reveals strong contributions along paths parallel to the Γ -X direction.

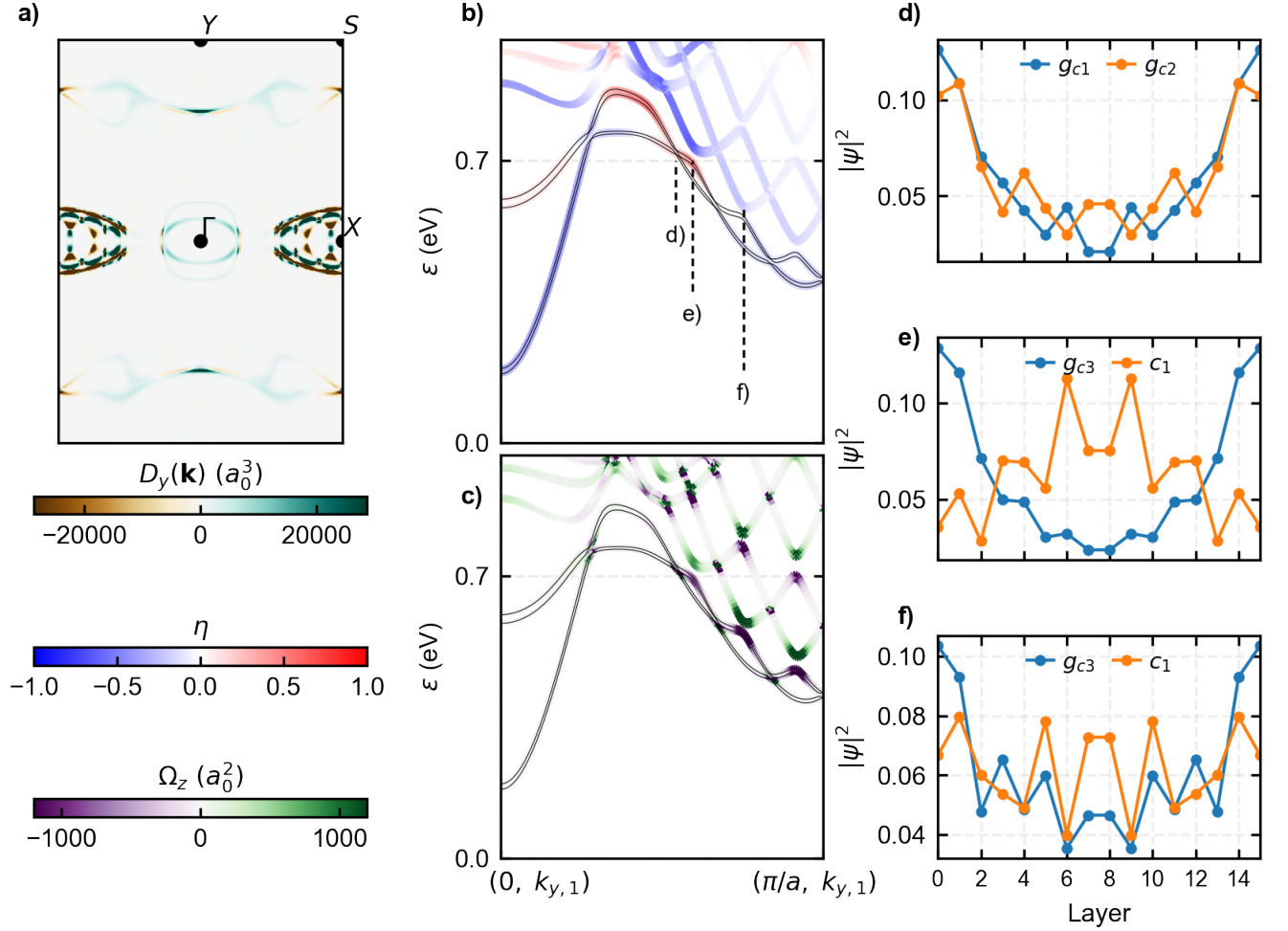


FIG. 6. Momentum-space and layer-resolved characterization of the avoided crossings in the $n = 8$ slab. (a) Momentum-resolved distribution of the Berry-curvature dipole density $D_y(\mathbf{k})$ for the $n = 8$ slab over the first Brillouin zone. The color scale is saturated at the 98th percentile of the data. (b) Band structure along the line from $(0, k_{y,1})$ to $(\pi/a, k_{y,1})$, with fixed $k_{y,1} = 0.03 a_0^{-1}$. Color encodes the surface polarization η ; the dashed vertical lines mark three representative avoided crossings. (c) Same band structure as in (b), with color indicating the Berry curvature Ω_z . (d–f) Layer-resolved probability weight of the Bloch states forming, respectively, the left, middle, and right avoided crossings highlighted in (b). The layer profiles reveal the spatial localization of the states involved in different avoided crossings.

The symmetry properties of $D_y(\mathbf{k})$ are consistent with those of the slab. First, time-reversal symmetry imposes $D_y(-\mathbf{k}) = D_y(\mathbf{k})$. Second, the mirror symmetry M_y requires $D_y(k_x, -k_y) = D_y(k_x, k_y)$. Combining these two constraints further yields $D_y(-k_x, k_y) = D_y(k_x, k_y)$.

The BCD hot spots identified in panel (a) coincide with localized enhancements of the Berry curvature associated with avoided crossings. To illustrate their microscopic origin, Fig. 6(b) shows the band structure along a representative path crossing these hot spots, $(0, k_{y,1}) \rightarrow (\pi/a, k_{y,1})$, with fixed $k_{y,1} = 0.03 a_0^{-1}$, with the bands colored according to their surface polarization $\eta(\mathbf{k})$. Figure 6(c) shows the same band structure colored by the Berry curvature $\Omega_z(\mathbf{k})$. The localized peaks in Ω_z occur at avoided crossings between states with different degrees of surface localization, three representative examples of

which are marked by the dashed lines in panel (b).

The layer-resolved wave-function weights shown in Fig. 6(d)–(f) clarify the nature of these avoided crossings. Panel (d) illustrates an avoided crossing between two states that remain predominantly localized near the surfaces. Such surface–surface hybridizations are also present in the unrelaxed slabs and therefore do not constitute a qualitatively new mechanism introduced by relaxation. Surface relaxation, however, substantially changes the situation by shifting surface-derived bands into the energy range occupied by bulk states. As a result, the relaxed slabs exhibit a proliferation of avoided crossings involving states with markedly different spatial character. Panels (e) and (f) show representative examples: in each case, a state with pronounced surface weight hybridizes with a state whose wave function

extends throughout the slab. These avoided crossings therefore correspond to surface–bulk hybridization.

This distinction is central to the enhancement of the BCD in the relaxed slabs. In the unrelaxed system, the relevant surface-derived states lie predominantly within the projected bulk gap, which restricts the number of bands with which they can hybridize. Surface relaxation shifts these states into the projected bulk continuum, where they become surface resonances and encounter a much denser set of bulk-like bands. The resulting proliferation of surface–bulk avoided crossings produces small interband energy separations and strong local enhancements of the Berry curvature. At the same time, a larger number of intermediate bands m contribute appreciably to the BCD of a given band n through Eq. (10). The combined increase in the number of relevant interband couplings and the reduction of their characteristic energy denominators generates the pronounced BCD hot spots visible in Fig. 6(a) and accounts for the overall enhancement of the BCD upon surface relaxation.

The enhancement of the Berry curvature associated with the interplay between surface and bulk states is not unique to the present system. In Weyl semimetals, Fermi arcs have been shown to exhibit a divergent surface Berry curvature as they merge into the bulk states [23]. Although GaAs does not host Fermi arcs and no such divergence occurs here, it is nevertheless noteworthy that an enhancement of the Berry curvature similarly emerges as surface-derived and bulk-like states approach each other in energy and hybridize.

V. CONCLUSIONS

We have investigated the microscopic origin of the Berry curvature dipole (BCD) in GaAs (110) slabs using density functional theory calculations combined with Wannier-function-based modeling. While the BCD vanishes in bulk GaAs due to symmetry constraints, the reduced symmetry at the surface allows for a finite dipole.

We have analyzed the differences between surfaces with and without structural relaxation, which is known to strongly affect the surface electronic structure of GaAs. In unrelaxed slabs, the BCD is mainly associated with exponentially localized surface states lying within the projected bulk band gap. Structural relaxation displaces the surface atoms, with As atoms shifting toward the vacuum and Ga atoms toward the interior of the slab. These structural changes shift the surface-derived states into the energy range of the bulk continuum and transform them into surface resonances. The resulting surface–

bulk hybridization generates avoided crossings and redistributes the Berry curvature in momentum space, leading to a substantial enhancement of the BCD, which reaches values approximately twice as large as those obtained for unrelaxed slabs.

Our analysis further reveals that the dominant contributions arise from interband processes involving surface-derived states and are strongly enhanced near avoided crossings and regions of small interband energy separation. The enhancement of the BCD can thus be understood as the combined effect of the relaxation-induced delocalization of the surface states and their increased coupling to bulk-like states. More broadly, our results highlight the role of surface resonances as a source of enhanced Berry curvature and nonlinear transport responses.

GaAs (110) surfaces, and potentially other members of the III–V semiconductor family, therefore provide an attractive experimental platform for exploring Berry-curvature-driven responses associated with surface electronic states. Since the strongest Berry-curvature features occur at energies beyond those readily accessible by conventional doping, our results also motivate the investigation of frequency-dependent nonlinear responses probing the same relaxation-induced surface–bulk hybridization [66–71].

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DATA AVAILABILITY

The data supporting the findings of this study are openly available [56]. The dataset includes the input files and converged charge densities used in the first-principles calculations, together with the information required to construct the Wannier models employed in this work.

CODE AVAILABILITY

The codes used to obtain the results presented in this article are available from the authors upon reasonable request.

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